Université de Lille

M2 Scientific Computing
2025/2026

Mathematical Tools for Simulation

Project

The C/C++ programming language is imposed.

The problem

We consider an advection-diffusion equation in the 2D square domain $\Omega := (0,1)^2$ over the time interval $[0,T]$, for a given simulation time $T > 0$. The boundary of the domain Ω is partitioned as $\partial\Omega = \overline{\Gamma^D} \cup \overline{\Gamma^N}$, where $\Gamma^D := \big(\{0\} \times (0,1)\big) \cup \big(\{1\} \times (0,1)\big)$ and $\Gamma^N := \big((0,1) \times \{0\}\big) \cup \big((0,1) \times \{1\}\big)$. The problem reads:

$$\left\{\begin{array}{rcll} \partial_t u + \operatorname{div}\big(-\boldsymbol{\nabla} u + \boldsymbol{V} u\big) & = & f & \text{in } (0,T] \times \Omega, \\ u & = & u^D & \text{on } (0,T] \times \Gamma^D, \\ \big(-\boldsymbol{\nabla} u + \boldsymbol{V} u\big) \cdot \boldsymbol{n} & = & 0 & \text{on } (0,T] \times \Gamma^N, \\ u(0,\cdot) & = & u^0 & \text{in } \Omega, \end{array}\right. \tag{1}$$

where $\boldsymbol{n} : \partial\Omega \to \mathbb{R}^2$ denotes the unit vector field, normal to $\partial\Omega$, pointing outward from Ω. Here, $\boldsymbol{V} : \Omega \to \mathbb{R}^2$ is a given advection field in $\boldsymbol{C}^1(\overline{\Omega})$ satisfying $\operatorname{div}(\boldsymbol{V}) = 0$ in Ω and $\boldsymbol{V} \cdot \boldsymbol{n} = 0$ on Γ^N. The functions $f : \Omega \to \mathbb{R}$, $u^D : \Gamma^D \to \mathbb{R}$, and $u^0 : \Omega \to \mathbb{R}$ respectively stand for the loading term, the Dirichlet boundary datum, and the initial datum. Notice that none of the previous data depends on the time variable.

To assess the performances of the different numerical schemes studied throughout this project, we will consider the following data, for some parameter $\eta \in \mathbb{R}$ enabling to tune the intensity of the advection: for all $\boldsymbol{x} := (x,y) \in \Omega$,

$$\boldsymbol{V}(\boldsymbol{x}) = (\eta, 0), \qquad f(\boldsymbol{x}) = 0, \qquad u^0(\boldsymbol{x}) = e^{\eta x} + e^{\frac{\eta}{2}x}\sin(\pi x), \tag{2}$$

and, for all $y \in (0,1)$, $u^D(0,y) = 1$ and $u^D(1,y) = e^{\eta}$.

Q0. Check that the above advection field does satisfy $\operatorname{div}(\boldsymbol{V}) = 0$ in Ω and $\boldsymbol{V} \cdot \boldsymbol{n} = 0$ on Γ^N. Check that

$$u(t,\boldsymbol{x}) = e^{\eta x} + e^{-\left(\pi^2 + \frac{\eta^2}{4}\right)t} e^{\frac{\eta}{2}x}\sin(\pi x)$$

solves Problem (1).

Part I: Finite Volume approximation

We consider a uniform Cartesian mesh of the domain Ω. For $k \in \mathbb{N}^\star$, we let $M := 2^k$ and $h := 1/M$, and we consider the mesh $(\mathcal{T}_h, \mathcal{E}_h)$ whose cells, collected in the set $\mathcal{T}_h$, are

$$K_{i,j} := (ih, (i+1)h) \times (jh, (j+1)h), \qquad i = 0, \dots, M-1,\ j = 0, \dots, M-1.$$

For any cell $K_{i,j}$ in $\mathcal{T}_h$, we denote by $\boldsymbol{x}_{i,j} \in \mathbb{R}^2$ its barycenter. The set $\mathcal{E}_h$ collects the edges of the mesh, divided into interior and boundary edges, respectively collected in $\mathcal{E}_h^{int}$ and $\mathcal{E}_h^{bnd}$. Boundary edges in $\mathcal{E}_h^{bnd}$ are themselves divided into Dirichlet and Neumann edges, respectively collected in $\mathcal{E}_h^D$ and $\mathcal{E}_h^N$. For an interior edge $\sigma \in \mathcal{E}_h^{int}$ such that $\sigma = K_{i,j} \mid K_{i',j'}$, we define

$$d_\sigma := |\boldsymbol{x}_{i',j'} - \boldsymbol{x}_{i,j}|.$$

For a boundary edge $\sigma \in \mathcal{E}_h^{bnd}$ such that $\sigma \subseteq \partial\Omega \cap \partial K_{i,j}$, denoting $\boldsymbol{x}_\sigma$ the midpoint of σ, we define

$$d_\sigma := |\boldsymbol{x}_{i,j} - \boldsymbol{x}_\sigma|.$$

For any cell $K_{i,j} \in \mathcal{T}_h$, we let $\mathcal{E}_{K_{i,j}}$ denote the collection of edges composing the boundary of $K_{i,j}$, and for any $\sigma \in \mathcal{E}_{K_{i,j}}$, we let $\boldsymbol{n}_{K_{i,j},\sigma}$ denote the unit vector normal to σ pointing outward from $K_{i,j}$. For the discretization of the Dirichlet boundary condition, we finally introduce, for any edge $\sigma \in \mathcal{E}_h^D$,

$$u_\sigma^D := u^D(\boldsymbol{x}_\sigma).$$

Q1. What is the value of d_σ if σ is an interior edge, or if σ is a boundary edge?

Q2. Create a class `Mesh`, itself based on objects of type `Cell`, `Edge`, and `Vertex`, formalizing the notion of a mesh. Implement in this class a function `build_Cartesian`, which takes k as an input, and builds the mesh object $(\mathcal{T}_h, \mathcal{E}_h)$ as defined just above. These structures will be reused in Part II of this project.

Let $N \in \mathbb{N}^\star$, and let $\Delta t := T/N$ be a (uniform) time step.

We consider a forward Euler time discretization of Problem (1), leading to the generic scheme:

$$\frac{|K_{i,j}|}{\Delta t}(u_{i,j}^{n+1} - u_{i,j}^n) + \sum_{\sigma \in \mathcal{E}_{K_{i,j}}} \mathcal{F}_{K_{i,j},\sigma}^n = |K_{i,j}| f_{i,j}. \tag{3}$$

We introduce the decomposition

$$\mathcal{F}_{K_{i,j},\sigma}^n := \mathcal{F}_{K_{i,j},\sigma}^{L,n} + \mathcal{F}_{K_{i,j},\sigma}^{T,n}, \tag{4}$$

where the superscripts L and T respectively stand for "Laplace" and "transport": the term $\mathcal{F}_{K_{i,j},\sigma}^{L,n}$ is obtained by discretization of $-\Delta u$, whereas $\mathcal{F}_{K_{i,j},\sigma}^{T,n}$ is obtained by discretization of $\mathrm{div}(\boldsymbol{V}u)$.

Q3. Justify how the generic scheme (3) is obtained. Precise the definitions of the $f_{i,j}$'s and of the $u_{i,j}^0$'s. Precise which "continuous" quantities $\mathcal{F}_{K_{i,j},\sigma}^{L,n}$ and $\mathcal{F}_{K_{i,j},\sigma}^{T,n}$ are meant to approximate.

We use a standard two-point scheme for $\mathcal{F}^{L,n}_{K_{i,j},\sigma}$, and an upwind scheme for $\mathcal{F}^{T,n}_{K_{i,j},\sigma}$. We let

$$V_{K_{i,j},\sigma} := \frac{1}{|\sigma|}\int_\sigma \boldsymbol{V}\cdot\boldsymbol{n}_{K_{i,j},\sigma}.$$

Q4. Express explicitly $\mathcal{F}^{L,n}_{K_{i,j},\sigma}$ and $\mathcal{F}^{T,n}_{K_{i,j},\sigma}$ as functions of some $u^n_{i',j'}$'s and u^D_σ's.

Q5. Write the scheme (3)–(4) under the form $u^{n+1}_{i,j} = L_{i,j}\left((u^n_{i',j'}),(u^D_\sigma)\right) + \Delta t f_{i,j}$, where $L_{i,j}$ is linear.

Q6. Give a condition on Δt, in terms of h and $\|\boldsymbol{V}\|_{\boldsymbol{L}^\infty(\Omega)}$, ensuring that $L_{i,j}\left((u^n_{i',j'}),(u^D_\sigma)\right)$ is a convex combination of the $u^n_{i',j'}$'s and u^D_σ's. If this condition is satisfied and $f\equiv 0$, what can you conclude about the behaviour of the scheme?

Q7. Based on **Q2**, implement the scheme (3)–(4), and perform a numerical study of its convergence, as h tends to zero, on test-case (2) for $\eta = 1$ and $T = 0.5$. This shall be done by plotting in log-log scale the error $\|\boldsymbol{u}^N - \boldsymbol{u}(T)\|_{h,2}$ at final time T as a function of h, with $\boldsymbol{u}^N := \left(u^N_{i,j}\right)$ and $\boldsymbol{u}(T) := \left(u(T,\boldsymbol{x}_{i,j})\right)$, where for any $\boldsymbol{v} := (v_{i,j}) \in \mathbb{R}^{M^2}$,

$$\|\boldsymbol{v}\|_{h,2} := \left(\sum_{i,j=0}^{M-1} |K_{i,j}||v_{i,j}|^2\right)^{1/2}.$$

We now consider a more general discretization of Problem (1). Let $B:\mathbb{R}\to\mathbb{R}$ satisfying

(a) $B(z)\geq 0$ for all $z\in\mathbb{R}$;
(b) $B(0)=1$;
(c) B is continuous and piecewise C^1;
(d) for all $z\in\mathbb{R}$, $B(-z)-B(z)=z$.

For the combined discretization of both the Laplace and transport terms, we set

$$\mathcal{F}^n_{K_{i,j},\sigma} := \begin{cases} \dfrac{|\sigma|}{d_\sigma}\left(B(-V_{K_{i,j},\sigma}\,d_\sigma)\,u^n_{i,j} \;-\; B(V_{K_{i,j},\sigma}\,d_\sigma)\,u^n_{i',j'}\right) & \text{for } \sigma := K_{i,j}\mid K_{i',j'}\in\mathcal{E}^{int}_h,\\ \dfrac{|\sigma|}{d_\sigma}\left(B(-V_{K_{i,j},\sigma}\,d_\sigma)\,u^n_{i,j} \;-\; B(V_{K_{i,j},\sigma}\,d_\sigma)\,u^D_\sigma\right) & \text{for } \sigma\subseteq\partial\Omega\cap\partial K_{i,j}\in\mathcal{E}^D_h,\\ 0 & \text{for } \sigma\subseteq\partial\Omega\cap\partial K_{i,j}\in\mathcal{E}^N_h. \end{cases} \tag{5}$$

Q8. Does the function defined by $B(z) = 1+\max(-z,0)$ satisfy the assumptions (a)–(d) (justify your answer)? Show that, with this function, (5) is equivalent to (4) with $\mathcal{F}^{L,n}_{K_{i,j},\sigma}$, $\mathcal{F}^{T,n}_{K_{i,j},\sigma}$ defined as in **Q4**.

We now introduce the so-called Bernoulli function:

$$B(z) := \begin{cases} \dfrac{z}{e^z-1} & \text{for all } z\in\mathbb{R}\backslash\{0\},\\ 1 & \text{if } z=0. \end{cases} \tag{6}$$

Q9. Show that this function complies with (a)–(d).

The scheme (3)–(5)–(6) is referred to as the Scharfetter–Gummel (SG) scheme in the literature.

Q10. Write the SG scheme under the form $u_{i,j}^{n+1} = \tilde{L}_{i,j}\left((u_{i',j'}^n), (u_\sigma^D)\right) + \Delta t f_{i,j}$, where $\tilde{L}_{i,j}$ is linear.

Q11. Give a condition on Δt, in terms of h and $\|\boldsymbol{V}\|_{\boldsymbol{L}^\infty(\Omega)}$, ensuring that $\tilde{L}_{i,j}\left((u_{i',j'}^n), (u_\sigma^D)\right)$ is a convex combination of the $u_{i',j'}^n$'s and u_σ^D's.

Q12. Implement the SG scheme and study, as it was done in **Q7**, its convergence in h on test-case (2) for $\eta = 1$ and $T = 0.5$.

The long-time behaviour of the advection-diffusion equation (1) is well-known: when $t \to \infty$, the solution $u(t, \cdot)$ to (1) converges exponentially fast towards some steady-state $u^\infty : \Omega \to \mathbb{R}$.

Q13. We consider test-case (2) for $\eta = 1$.

a) Give the explicit expression of u^∞.

b) For a given value of h (and appropriate Δt), perform a simulation for $T = 5$ with the schemes (3)–(4) (upwind) and (3)–(5)–(6) (SG), stopping the time loop when $\|\boldsymbol{u}^{n+1} - \boldsymbol{u}^n\|_{h,2}$ does not evolve anymore (up to a tolerance of 10^{-13}). Denoting by n_∞ the stopping time index (which may be different for the two schemes), we shall define the discrete steady-state of each scheme by $\boldsymbol{u}^{n_\infty}$.

c) We let $\boldsymbol{u}^\infty := \left(u^\infty(\boldsymbol{x}_{i,j})\right)$. For each scheme, plot in semilog-y scale the time evolution (i.e. with respect to n) of $\|\boldsymbol{u}^n - \boldsymbol{u}^\infty\|_{h,2}$ on the one hand, and of $\|\boldsymbol{u}^n - \boldsymbol{u}^{n_\infty}\|_{h,2}$ on the other hand. Comment. Does the result change if you diminish the value of h in the simulations of **b)**?

Part II: Finite Element approximation

We recall that the advection field satisfies $\mathrm{div}(\boldsymbol{V}) = 0$ in Ω and $\boldsymbol{V} \cdot \boldsymbol{n} = 0$ on Γ^N, and that neither $\boldsymbol{V}$, nor the source term f, nor the Dirichlet datum u^D depend on the time variable.

We henceforth assume that $f, u^0 \in L^2(\Omega)$, and that $u^D \in H^{1/2}(\Gamma^D)$. We will admit that there exists a lifting $\tilde{u}^D \in H^1(\Omega)$ such that $\gamma(\tilde{u}^D)_{|\Gamma^D} = u^D$, and satisfying $\|\tilde{u}^D\|_{H^1(\Omega)} \le C_{L,D}\|u^D\|_{H^{1/2}(\Gamma^D)}$.

Q14. Write a semi-variational (in space only) formulation for Problem (1) under the form: Find $u : (0,T) \to H^1(\Omega)$, $t \mapsto u(t) := \tilde{u}(t) + \tilde{u}^D$, with $\tilde{u}(t) \in H_0^{1,D}(\Omega)$ such that

$$\begin{cases} \dfrac{\mathrm{d}}{\mathrm{d}t}\left(\displaystyle\int_\Omega \tilde{u}(t)\, v\right) + a\big(\tilde{u}(t), v\big) = \displaystyle\int_\Omega f v - a\big(\tilde{u}^D, v\big) \qquad \forall\, v \in H_0^{1,D}(\Omega), \quad \text{for a.e. } t \in (0,T), \\ \tilde{u}(0) = u^0 - \tilde{u}^D, \end{cases} \tag{7}$$

where the space $H_0^{1,D}(\Omega)$ (endowed with the norm $\|\boldsymbol{\nabla} v\|_{\boldsymbol{L}^2(\Omega)}$) shall be given a precise definition and proven to be a Hilbert space, and the bilinear form $a : H^1(\Omega) \times H^1(\Omega) \to \mathbb{R}$ shall be made explicit.

Q15. Consider the steady problem: Find $H^1(\Omega) \ni u^\infty := \tilde{u}^\infty + \tilde{u}^D$, with $\tilde{u}^\infty \in H_0^{1,D}(\Omega)$ such that

$$a\big(\tilde{u}^\infty, v\big) = \int_\Omega f v - a\big(\tilde{u}^D, v\big) \qquad \forall\, v \in H_0^{1,D}(\Omega). \tag{8}$$

Prove that this problem is well-posed in the Hadamard sense. It might be helpful to use the fact that $\mathrm{div}(\boldsymbol{V}) = 0$ in Ω and that $\boldsymbol{V} \cdot \boldsymbol{n} = 0$ on Γ^N.

Q16. Remarking that $\big(u(t) - u^\infty\big) = \big(\tilde{u}(t) - \tilde{u}^\infty\big) \in H_0^{1,D}(\Omega)$, and comparing (that is, subtracting) the two problems (7) and (8), prove that

$$\|u(t) - u^\infty\|_{L^2(\Omega)} \leq e^{-\beta t} \|u^0 - u^\infty\|_{L^2(\Omega)},$$

with a constant $\beta > 0$, only depending on the domain Ω, to be made precise. It might be helpful to invoke the Grönwall lemma (in its differential form). Conclude on the long-time behaviour of the solution to (1).

We now study the conforming finite element approximation of the steady Problem (8) in some finite-dimensional space $U_h \subset H^1(\Omega) \cap C^0(\overline{\Omega})$ on the uniform Cartesian mesh $(\mathcal{T}_h, \mathcal{E}_h)$ (cf. Part I). Notice that $\mathrm{diam}\big(K_{i,j}\big) = \sqrt{2}\, h$ for all $0 \leq i, j \leq M - 1$. We henceforth assume that $\tilde{u}^D \in U_h$, and we let $U_{h,0}^D$ denote the subspace of U_h such that $U_{h,0}^D \subset H_0^{1,D}(\Omega)$. We seek an approximation of u^∞ under the form $U_h \ni u_h^\infty := \tilde{u}_h^\infty + \tilde{u}^D$, with $\tilde{u}_h^\infty \in U_{h,0}^D$ satisfying

$$a\big(\tilde{u}_h^\infty, v_h\big) = \int_\Omega f v_h - a\big(\tilde{u}^D, v_h\big) \qquad \forall\, v_h \in U_{h,0}^D. \tag{9}$$

Q17. Remarking that $\big(u^\infty - u_h^\infty\big) = \big(\tilde{u}^\infty - \tilde{u}_h^\infty\big) \in H_0^{1,D}(\Omega)$, and revisiting the proof of the Céa's lemma, show that

$$\|\boldsymbol{\nabla}\big(u^\infty - u_h^\infty\big)\|_{\boldsymbol{L}^2(\Omega)} \leq \inf_{v_h \in U_h \,|\, \gamma(v_h)_{|\Gamma^D} = u^D} \big(\|\boldsymbol{\nabla}\big(u^\infty - v_h\big)\|_{\boldsymbol{L}^2(\Omega)} + \|\boldsymbol{V}\|_{\boldsymbol{L}^\infty(\Omega)} \|u^\infty - v_h\|_{L^2(\Omega)}\big). \tag{10}$$

Q18. Assume that $u^\infty \in H^2(\Omega)$, and that there exists an interpolation operator $\mathcal{I}_h : H^1(\Omega) \cap C^0(\overline{\Omega}) \to U_h$, satisfying $\mathcal{I}_h(v_h) = v_h$ for all $v_h \in U_h$, $\mathcal{I}_h\big(H_0^{1,D}(\Omega) \cap C^0(\overline{\Omega})\big) \subseteq U_{h,0}^D$, and such that

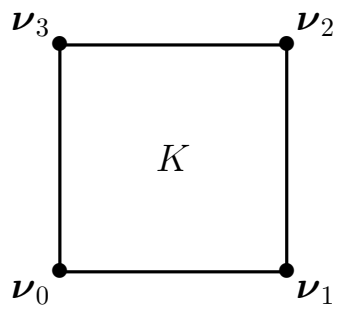
$$\|\boldsymbol{\nabla}\big(v - \mathcal{I}_h(v)\big)\|_{\boldsymbol{L}^2(\Omega)} \leq C\, h\, |v|_{H^2(\Omega)}, \qquad \|v - \mathcal{I}_h(v)\|_{L^2(\Omega)} \leq C'\, h^2\, |v|_{H^2(\Omega)} \qquad \forall v \in H^2(\Omega),$$

for two constants $C, C' > 0$ independent of h. Starting from the abstract estimate (10), establish an explicit error estimate on $\|\boldsymbol{\nabla}\big(u^\infty - u_h^\infty\big)\|_{\boldsymbol{L}^2(\Omega)}$, and discuss its validity depending on the value of the so-called Péclet number $\mathrm{Pe} := h\|\boldsymbol{V}\|_{\boldsymbol{L}^\infty(\Omega)}$. Comment.

We consider the (conforming) $\mathbb{Q}^1$-FE approximation of the steady Problem (8) on the mesh $(\mathcal{T}_h, \mathcal{E}_h)$. We let

$$\mathbb{Q}^1 := \big\{\alpha + \beta_1\, x + \beta_2\, y + \gamma\, xy,\ (\alpha, \beta_1, \beta_2, \gamma) \in \mathbb{R}^4\big\},$$

and we denote by $(\boldsymbol{\nu}_\ell \in \mathbb{R}^2)_{0 \leq \ell \leq 3}$ the positions of the 4 vertices of any generic square $K \in \mathcal{T}_h$.



Q19. Prove that the triple $\big(K, \mathbb{Q}^1(K), \{\tau_\ell\}_{0\le\ell\le 3}\big)$, with $\tau_\ell : \mathbb{Q}^1(K) \to \mathbb{R}$ such that $\tau_\ell(v) := v(\boldsymbol{\nu}_\ell)$ for all $v \in \mathbb{Q}^1(K)$ and $0 \le \ell \le 3$, is unisolvent as soon as the square K is non-degenerate.

Q20. Verify that, for a given square $K \in \mathcal{T}_h$, the local shape functions for the $\mathbb{Q}^1$-FE are given, for $0 \le \ell \le 3$, by

$$\varphi_\ell(\boldsymbol{x}) := \frac{\big(x - \nu_{(\ell+2)[4],x}\big)\big(y - \nu_{(\ell+2)[4],y}\big)}{\big(\nu_{\ell,x} - \nu_{(\ell+2)[4],x}\big)\big(\nu_{\ell,y} - \nu_{(\ell+2)[4],y}\big)} \qquad \forall\, \boldsymbol{x} := (x,y) \in \overline{K}.$$

We henceforth define

$$U_h := \big\{v_h \in C^0(\overline{\Omega}) \mid v_{h|K} \in \mathbb{Q}^1(K)\ \forall K \in \mathcal{T}_h\big\} \quad \text{and} \quad U_{h,0}^D := \big\{v_h \in U_h \mid \gamma(v_h)_{|\Gamma^D} \equiv 0\big\}. \tag{11}$$

The interpolation operator $\mathcal{I}_h$ is chosen to be the canonical $\mathbb{Q}^1$-FE interpolant, satisfying, for v smooth enough, $[\mathcal{I}_h(v)](\boldsymbol{\nu}) = v(\boldsymbol{\nu})$ for all vertices $\boldsymbol{\nu}$ of the mesh $(\mathcal{T}_h, \mathcal{E}_h)$.

Q21. Justify that a function v defined on $\overline{K} \cup \overline{L}$, with $K, L \in \mathcal{T}_h$ sharing the edge $\sigma := (\boldsymbol{\nu}_\flat, \boldsymbol{\nu}_\sharp) \in \mathcal{E}_h^{int}$, such that $v_{|K} \in \mathbb{Q}^1(K)$, $v_{|L} \in \mathbb{Q}^1(L)$, and $v_{|K}(\boldsymbol{\nu}_\flat) = v_{|L}(\boldsymbol{\nu}_\flat)$, $v_{|K}(\boldsymbol{\nu}_\sharp) = v_{|L}(\boldsymbol{\nu}_\sharp)$, satisfies $v \in C^0\big(\overline{K} \cup \overline{L}\big)$.

Q22. Justify that, for test-case (2), it is indeed possible to construct a lifting $\tilde{u}^D \in U_h$ of the Dirichlet datum u^D. In practice, this lifting will be taken such that $\tilde{u}^D(\boldsymbol{\nu}) = 0$ for all vertices $\boldsymbol{\nu}$ not pertaining to $\overline{\Gamma^D}$.

Q23. Based on **Q2**, implement the discrete steady Problem (9)–(11), and perform a numerical study of its convergence (plot in log-log scale the errors $\|\boldsymbol{\nabla}\big(\mathcal{I}_h(u^\infty) - u_h^\infty\big)\|_{\boldsymbol{L}^2(\Omega)}$ and $\|\mathcal{I}_h(u^\infty) - u_h^\infty\|_{L^2(\Omega)}$ as functions of h) on the (steady) test-case (2) for $\eta = 1$, $\eta = 10$, and $\eta = 100$ (see **Q13**, **a)** for the corresponding exact solution). What do you remark? What link can you draw with **Q18**?

<u>BONUS:</u> We consider a backward Euler time discretization, coupled with a (conforming) $\mathbb{Q}^1$-FE space discretization of Problem (7), leading to the following fully discrete problem:
Find $\big(u_h^n := \tilde{u}_h^n + \tilde{u}^D\big)_{1\le n\le N} \in \big(U_h\big)^N$, with $\big(\tilde{u}_h^n\big)_{1\le n\le N} \in \big(U_{h,0}^D\big)^N$ such that

$$\begin{cases} \displaystyle\int_\Omega \left(\frac{\tilde{u}_h^n - \tilde{u}_h^{n-1}}{\Delta t}\right) v_h + a\big(\tilde{u}_h^n, v_h\big) = \int_\Omega f v_h - a\big(\tilde{u}^D, v_h\big) \qquad \forall\, v_h \in U_{h,0}^D, \quad \forall\, 1 \le n \le N, \\ \tilde{u}_h^0 = u_h^0 - \tilde{u}^D, \end{cases} \tag{12}$$

where $u_h^0 \in U_h$ is chosen to be $\mathcal{I}_h(u^0)$ (u^0 is assumed smooth enough).

Q24. Is the stability of Problem (12) dependent on some CFL (smallness) condition on Δt?

Q25. Remarking that $\big(u_h^n - u_h^\infty\big) = \big(\tilde{u}_h^n - \tilde{u}_h^\infty\big) \in U_{h,0}^D$ for $n \ge 1$, and comparing the two discrete problems (12) and (9), prove a discrete equivalent on $\|u_h^n - u_h^\infty\|_{L^2(\Omega)}$ of the result of **Q16**. It might be helpful to use the fact that, for all $a, b \in \mathbb{R}$, $(a-b)a \ge \frac{a^2-b^2}{2}$. Conclude about the long-time behaviour of the (discrete) solution to (12).

Q26. We consider test-case (2) for $\eta = 1$. For given values of h and Δt, perform a simulation with the scheme (12) for $T = 5$, and plot in semilog-y scale the time evolution (i.e. with respect to n) of the quantity $\|u_h^n - \mathcal{I}_h(u^\infty)\|_{L^2(\Omega)}$ on the one hand, and of the quantity $\|u_h^n - u_h^\infty\|_{L^2(\Omega)}$ on the other hand. Test for smaller values of h. Comment.